Secure Cloud Document Management & Collaboration Platform

Feature Overview Document

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Solution: Nextcloud on Amazon Web Services (AWS) — Mumbai Region
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Quick Glossary — Technology Terms Used In This Document

Since this document mentions a few technology terms, a short glossary is provided upfront for easy reference.

Term	Meaning in Plain Language
Cloud	Storing your files and running your software on secure servers over the internet, instead of on physical hard drives kept in your office.
AWS (Amazon Web Services)	The world's largest cloud infrastructure provider. Your files will physically live on their servers in **Mumbai**.
Nextcloud	A well-established open-source software platform that turns cloud storage into a private "Google Drive" or "Dropbox" for your firm — with file editing, sharing, chat, and audit trails built in.
Server	A powerful computer running 24×7 that hosts your Nextcloud portal.
Storage (S3)	The cloud location where all your firm's files physically reside. Highly durable — Amazon guarantees 99.999999999% data durability.
Database (RDS / MySQL)	A small structured record of *who has access to what file* and *what happened when*. Files themselves are stored separately.

Term	Meaning in Plain Language
Domain / URL	The web address your team will use (e.g. `files.yourfirm.co.in`).
HTTPS / TLS	The "lock symbol" in your browser — meaning all data traveling between users and the server is encrypted.
Encryption at rest	Files are stored in scrambled form on the server. Even if someone stole the hard drive, they could not read the data.
Backup	A separate, safely-stored copy of your data used to restore in case of accidental deletion or emergency.
Audit log	An automatic record of every action — who viewed, edited, downloaded, or deleted each file, and when.
Reserved Instance	A commitment to use a specific server for 1 year in exchange for a ~30% discount from Amazon.

About The Solution

The proposed platform gives your firm a **secure, private cloud-based document management and collaboration system**, entirely hosted on Amazon Web Services in the Mumbai region. It combines two mature, trusted technologies:

1. **Nextcloud** — an open-source enterprise document platform used by 400+ million users worldwide, including government agencies and large corporations.
2. **Amazon Web Services (AWS)** — the world's largest cloud provider, holding ISO 27001, SOC 2, and PCI-DSS certifications, with a dedicated India (Mumbai) region.

Together, they deliver a private "your-own-Drive" for the firm — files, editing, sharing, backup, and audit — with your data staying inside India, on infrastructure owned by your firm.

Please find the features proposed below:

Section A — File Storage & Management

#	Feature	Description
A1	Unlimited folder structure	Create folders, sub-folders and sub-sub-folders to any depth.
A2	Drag-and-drop upload	Drag files or entire folders from your PC into the browser.
A3	Chunked uploads	Large files (up to 16 GB per file) upload reliably in pieces.
A4	Resume-on-failure	Interrupted uploads automatically continue from where they stopped.
A5	Bulk download as ZIP	Download an entire client folder as a single compressed archive.
A6	In-browser file preview	Instantly view PDFs, images, videos, Office files without downloading.
A7	Thumbnails	Image and video thumbnails displayed in every folder view.
A8	Full-text search	Search inside PDFs, Word and Excel files — not just filenames.
A9	File tags	Apply labels to any file or folder for quick filtering.
A10	Favourites (starred items)	Pin frequently-used folders to the sidebar.
A11	Comments on files	Discussion thread attached to any file.
A12	File-level activity log	Right-side panel shows every action ever performed on that file.
A13	Metadata sidebar	Size, dates, creator, permissions, comments, tags in one view.
A14	External storage mounts	Ability to attach external SMB / S3 / WebDAV drives if ever needed.
A15	File locking	Prevent concurrent edits on files that must not be modified simultaneously.

#	Feature	Description
A16	Move / copy files across shared folders	Standard right-click operations.
A17	Sort by name / size / date / tag	Multiple sort orders in every folder.
A18	Grid or list view	User's choice of layout.

Section B — Document Editing (Nextcloud Office / Collabora)

#	Feature	Description
B1	Edit Word (.docx) in browser	Full Microsoft-Word-compatible editor.
B2	Edit Excel (.xlsx) in browser	Formulas, charts, pivot tables, data validation.
B3	Edit PowerPoint (.pptx) in browser	Slide transitions, animations, notes.
B4	Open Office formats	Native support for ODT, ODS, ODP.
B5	Real-time co-authoring	Two or more users edit the same document simultaneously, seeing each other's cursors.
B6	Track changes	Word-style redlining and change acceptance.
B7	Comments in documents	Cell-level or paragraph-level threaded discussions.
B8	Export to PDF	Convert any document to PDF with one click.
B9	Templates library	Pre-loaded firm letterhead, notice templates, engagement letters etc.
B10	Print any document	Direct printing from the browser.
B11	Save-as another format	Convert between DOCX ↔ ODT, XLSX ↔ ODS, etc.
B12	Version history within editor	See who edited what and when, restore earlier versions.
B13	Digital signatures on PDF (LibreSign)	Draw, upload or use a digital certificate to sign PDFs inside Nextcloud.
B14	PDF form filling	Fill interactive PDF forms directly in the browser.

Section C — Sharing & Collaboration

#	Feature	Description
C1	Share with internal users	Grant view or edit rights to any staff member with a single click.
C2	Share with a group	e.g. share a client folder with the "Firm-Partners" group.
C3	Public shareable link	Generate a link a client can open in any browser.
C4	Link expiry date	Auto-revoke access after a chosen date.
C5	Link password protection	Recipient must enter a password to open.
C6	Disable download option	Recipients can view but not download the file.
C7	Editable public link	Allow external auditors to edit if allowed.
C8	File-request link	Upload-only link — external party can drop files but cannot see other files.
C9	Share notifications	Email + in-app alert when a shared link is accessed.
C10	Instant revoke	Remove access from any share with one click.
C11	Share summary dashboard	Single view of all files you have shared, with expiry and status.

Section D — Team Communication (Nextcloud Talk)

#	Feature	Description
D1	1-to-1 text chat	Persistent, searchable conversations.
D2	Group chat rooms	For teams, client engagements, or internal projects.
D3	Voice calls	Encrypted 1-to-1 audio calls between staff.
D4	Video calls	Up to 4 people out-of-the-box; more with an optional add-on.

#	Feature	Description
D5	Screen sharing	Share your screen during a call.
D6	File drops in chat	Attach files directly to messages.
D7	Emoji + message reactions	Standard modern chat conventions.
D8	Voice messages	Record and send audio clips.
D9	Message replies + threads	Structured conversations.
D10	Public chat rooms via link	External clients can join a specific conversation.
D11	Chat search	Full-text across all messages.
D12	Meeting scheduler	Integrates with Calendar to create video rooms with reminders.
D13	Mobile push notifications	Instant notifications on iOS and Android.
D14	Waiting rooms	Meeting host approves each participant before entry.
D15	Kick / mute controls	Meeting host retains full control over participants.

Section E — Calendar, Contacts, Tasks (Groupware)

#	Feature	Description
E1	Personal calendar	Every user gets a personal calendar.
E2	Shared team calendars	e.g. "Firm-Deadlines" visible to all staff.
E3	Public read-only calendars	Statutory deadlines, MCA due-dates.
E4	Recurring events	Weekly, monthly, yearly recurrences with exceptions.
E5	Event reminders	Email + push notification before events.
E6	CalDAV synchronisation	Sync with Outlook, Apple Calendar, Thunderbird, mobile.
E7	Meeting scheduler	Find a free slot across multiple attendees.
E8	Attach files to events	Link agenda documents or supporting files.

#	Feature	Description
E9	Auto-create Talk room from event	One-click to convert a scheduled meeting into a video call room.
E10	Shared address book	Firm-wide contact directory.
E11	CardDAV synchronisation	Contacts sync to phones, Outlook, mobile apps.
E12	vCard import / export	Bring existing contacts in, share externally.
E13	Tasks / to-do lists	Personal and shared task lists, sync to mobile.
E14	Task categories, due dates, priorities	Full task management.

Section F — Task & Work Tracking (Deck + Time Manager)

#	Feature	Description
F1	Kanban boards	Visual work tracking (To-Do → In Progress → Done).
F2	Cards with title and description	Represent each piece of client work.
F3	Assign cards to team members	See who owns what.
F4	Due dates + reminders	Track deadlines per assignment.
F5	Labels / tags on cards	e.g. "Urgent", "AF-24-25", "Certified".
F6	Card comments	Team discussion per work item.
F7	Card attachments (linked to Nextcloud files)	Link supporting documents directly.
F8	Sub-tasks / checklists	Break work down into steps.
F9	Board sharing	Per user / group visibility.

Section G — Access from Any Device

#	Feature	Description
G1	Web browser access	Chrome / Edge / Firefox / Safari — no software installation required.
G2	Windows desktop sync client	The whole document library appears as a normal Windows folder.
G3	macOS desktop sync client	Same experience on Mac.
G4	Linux desktop sync client	For Linux users.
G5	Android mobile app	Free, on Play Store — full read/edit/upload.
G6	iOS mobile app	Free, on App Store — full read/edit/upload.
G7	Auto-photo-upload from phone	Take a photo of a document → auto-upload to a chosen folder.
G8	Offline mode (mobile)	Pin files to be available even without internet.
G9	On-demand file streaming	Virtual files that download only when opened — saves disk space on laptop.
G10	Selective sync	Choose which folders to sync to each device.
G11	Bandwidth throttling	Limit sync speed during working hours.
G12	WebDAV standard access	Compatible with third-party apps like Cyberduck, FileZilla, WinSCP.

Section H — Security

#	Feature	Description
H1	HTTPS / TLS 1.3 encryption in transit	All traffic between users and the server is encrypted end-to-end.
H2	AES-256 encryption at rest	Files stored on AWS are scrambled using bank-grade AES-256 encryption.

#	Feature	Description
H3	End-to-end encrypted folders (optional)	For most sensitive folders — even the server cannot read the contents.
H4	Password policy enforcement	Minimum length, complexity, no common passwords.
H5	Brute-force protection	Automatic throttling of failed logins to stop attackers.
H6	IP whitelist / blacklist	Optional restriction of login origin.
H7	Session timeout	Auto-logout after configurable idle period.
H8	Rate limiting	API and login abuse prevention.
H9	Trusted domains configuration	Prevents host-header attacks.
H10	Content Security Policy (CSP)	Cross-site-scripting protection.
H11	Anti-virus for uploads (ClamAV)	Server-side scanning of every uploaded file.
H12	Data centre security (AWS)	ISO 27001, SOC 1/2/3, PCI-DSS, HIPAA-ready certifications.
H13	AWS-side network hardening	Private network, security groups, no public database exposure.
H14	Server hardening	Ubuntu with automatic security patches, fail2ban, firewall.

Section I — Backup, Recovery & Version Control (three safety nets)

#	Feature	Description
I1	Trash bin (per user)	Deleted files remain recoverable for 30 days (configurable up to indefinite).
I2	File version history	Every edit automatically preserves the previous version.
I3	Version restore (self-service)	Users restore old versions from the file sidebar.
I4	Bulk restore from trash	Restore many files in one action.

#	Feature	Description
I5	Admin over-ride restore	Firm administrator can restore even after a user empties their trash.
I6	AWS S3 object versioning	Every past version of every file is retained in cloud storage.
I7	Daily EBS snapshots	Full server disk backed up every 24 hours.
I8	Daily RDS snapshots	Database backed up every 24 hours, restore to any point in the last 7 days.
I9	Glacier Deep Archive	Monthly full backup kept for 5–10 years for regulatory compliance.
I10	Cross-availability-zone backup	Backups stored in a separate AWS location for disaster resilience.
I11	Ransomware protection	Immutable versioned backups cannot be overwritten by an attacker.

Section J — Audit Log & Compliance

#	Feature	Description
J1	Every login recorded	User, IP address, device, browser, timestamp.
J2	Every failed login recorded	For security review.
J3	Every file event recorded	Viewed, downloaded, edited, renamed, moved, shared, deleted, restored.
J4	Every share event recorded	Created, accessed, expired, revoked.
J5	Every administrative action recorded	User creation, permission change, impersonation.
J6	Filter log by user	See everything a specific person did.
J7	Filter log by date range	e.g. "everything in April 2026".
J8	Filter log by action type	e.g. only "download" events.
J9	Filter log by file / folder	Full history of one file or folder.

#	Feature	Description
J10	CSV export	Take reports to the audit / management committee.
J11	AWS CloudTrail (infrastructure layer)	Every AWS API call independently logged.
J12	Data event logging on cloud storage	Object-level access recorded in CloudTrail.
J13	Log retention	Indefinite retention (configured to keep permanently).

Section K — User & Access Management (Admin Panel)

#	Feature	Description
K1	Create users individually	Standard onboarding.
K2	Bulk-create users via CSV	For quick trainee-batch onboarding.
K3	Group management	Partners / Seniors / Juniors / Trainees / Departed.
K4	Per-user storage quota	Set individual limits (e.g. 20 GB per user).
K5	Instant enable / disable user	One-click revoke access when a trainee leaves.
K6	Password reset from admin	Force reset if password compromised.
K7	Sub-admin roles	Delegate limited admin rights for HR / senior partner.
K8	Impersonation	Admin can log in as a user for troubleshooting (audited).
K9	Per-folder permission matrix	Assign different rights per folder per group.
K10	App management	Enable / disable Nextcloud apps as needs change.
K11	System health check dashboard	Warns of security or performance issues.
K12	Log viewer in admin UI	View server logs without needing technical shell access.

Section L — Data Residency & Ownership

#	Feature	Description
L1	All data physically stored in India	AWS Mumbai region (ap-south-1).
L2	Firm owns the AWS account	You control the master credentials.
L3	Firm owns the cloud storage contents	Downloadable at any time, no vendor lock-in.
L4	Full data export	Any user can export their entire data as a ZIP archive.
L5	Standard file formats retained	Word/Excel/PDF stored as-is, not proprietary containers.

Section M — Customisation & Branding

#	Feature	Description
M1	Custom firm logo	Displayed on login screen and top-left of every page.
M2	Custom brand colours	Match your firm's visual identity.
M3	Custom login page	Background image, welcome text, footer disclaimer.
M4	Dark and light mode	User-selectable theme.
M5	Multi-language interface	150+ languages including Hindi, Tamil, English.
M6	Configurable dashboard	Users pick which widgets appear on their home screen.

Section N — Notifications & Communication

#	Feature	Description
N1	In-app notifications	Bell icon top-right.
N2	Email notifications	Configurable digest or per-event.
N3	Mobile push notifications	For Talk messages, comments, mentions.
N4	Web push notifications	For browser-only users.
N5	@mention users in comments	They receive a direct notification.
N6	Notification preferences per user	Users control what they get.

Section O — Automation & Rules

#	Feature	Description
O1	Auto-tagging rules	e.g. auto-tag anything in `/2025-26/` as `Current FY`.
O2	Auto-moving files by rule	Route uploaded files to correct folder based on filename.
O3	Retention policies	Auto-delete working files after N days (if needed).
O4	File-request folders	Upload-only public folders for external clients.
O5	Command-line automation	Scriptable batch operations for admin.
O6	Background job scheduler	Cron-based system tasks.

Summary

The proposed platform delivers **130+ distinct features** across 15 categories — covering document storage, editing, collaboration, communication, security, backup, audit and administration. Everything a modern professional firm needs for its practice, while keeping full data ownership and India-resident hosting.

Reference — Official Documentation Links

Nextcloud

• Full documentation: https://docs.nextcloud.com/
• Feature tour: https://nextcloud.com/tour/
• Interactive demo (no signup): https://try.nextcloud.com/
• Nextcloud Talk features: https://nextcloud.com/talk/
• Nextcloud Office features: https://nextcloud.com/office/

AWS

• AWS Mumbai region: https://aws.amazon.com/local/india/
• AWS S3 (cloud storage): https://docs.aws.amazon.com/s3/
• AWS EC2 (cloud server): https://docs.aws.amazon.com/ec2/
• AWS Backup: https://docs.aws.amazon.com/aws-backup/
• AWS security & compliance: https://aws.amazon.com/compliance/

Document prepared by Logesh P S. For any clarifications, please reach out at any time.